

Project Outline

Form groups of 2 students each.

1. Dataset Selection and Understanding

- Choose a dataset from **Kaggle** that interests you.
- The dataset should:
 - Have **diverse features** (numerical & categorical).
 - Be **large enough** for meaningful analysis.
 - Have a **real-world application** (e.g., health, environment, business, sports, etc.).

✅ Examples of datasets:

- **Health:** Diabetes trends, heart disease predictors, mental health surveys.
- **Finance:** Stock prices, cryptocurrency trends, loan default prediction.
- **Sports:** FIFA player stats, Olympic medalists, NBA game results.
- **Environment:** Air quality, climate change, forest fires.
- **E-commerce:** Customer purchases, product reviews, sales trends.

2. Data Cleaning

- Handle **missing values** using imputation or removal techniques.
- Address **outliers** using statistical methods (e.g., Z-score, IQR).
- Convert **categorical variables** into numerical formats if needed.
- Normalize/standardize numerical data where necessary.

3. Text Data Encoding (If applicable)

- Encode categorical data (e.g., one-hot encoding for product categories).

4. Balance the dataset (If needed)

- Perform Oversampling or Undersampling if needed.

5. Exploratory Data Analysis & Hypothesis Generation

- Identify key **patterns and trends** in the dataset.
- Formulate a **hypothesis** (e.g., "Higher customer ratings lead to more sales").

- Each student should explore at least 2 different aspect of the hypothesis (e.g. "Visualize ratings though the seasons" and " explore the age of the customer that gave the higher ratings").
- Create **basic visualizations** to explore relationships:
 - ✓ Correlation heatmaps
 - ✓ Histograms & box plots for distributions
 - ✓ Scatter plots for relationships
- Perform Dimensionality Reduction to visualize your data if needed.

6. Data Visualization

- ✓ Display **key statistics** in an easy-to-read format.
- ✓ Include **multiple visualizations** (e.g., scatter plots, bar charts, time series).
- ✓ Allows **user interaction** (e.g., dropdowns, filters, sliders) (optional).

7. Insights & Recommendations

- Summarize key **findings and trends** from the dataset.
- Provide **data-driven recommendations** based on analysis.
- Discuss **challenges faced** and potential future improvements.

8. Oral Presentation

- Prepare the slides for an oral presentation
- The presentation should be approximately 15 minutes long.